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Machine Intelligence
Fall 2025

Exercise Sheet 3

1 Finite Domain Representation

1.1. Consider the **Robot and Keys** problem depicted in Figure 1. There is a robot that needs to pick up the key at the end of the corridor. The robot can move only one cell at the time, but it cannot step on the cell with locked padlock. It can pick up keys when in the same cell, and it can unlock the padlock when standing in the adjecent cell and has the corresponding key. The initial state is as depicted, the goal is to hold the key B in the minimum number of steps.

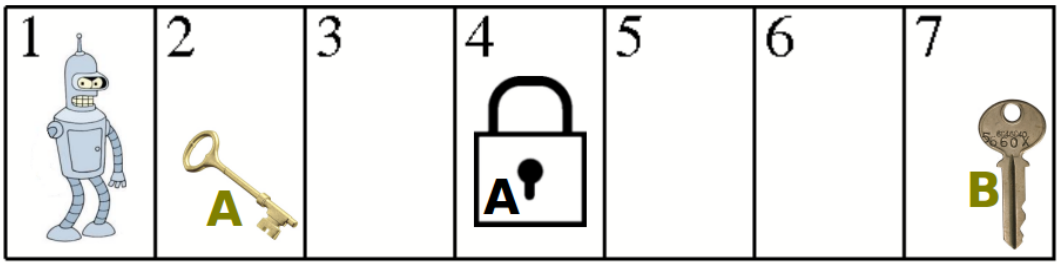

Figure 1: Illustration of the Robot and Keys problem

Model this problem as an FDR planning task.

Solution:

Variables:

- R, $D_R = \{1, ..., 7\}$ – position of the robot
- K_A, $D_{K_A} = \{0, 1\}$ – has key 1 or not
- K_B, $D_{K_B} = \{0, 1\}$ – has key 2 or not
- P, $D_P = \{0, 1\}$ – is padlock locked or not

Initial state: $\{R{=}1, K_A{=}0, K_B{=}0, P{=}1\}$

Goal: $\{K_B{=}1\}$

Actions:

- $move(x, y)$ – move from x to y, only for free cells;
  $(x, y) \in \{(1, 2), (2, 1), (2, 3), (3, 2), (5, 6), (6, 5), (6, 7), (7, 6)\}$
  - pre: $R = x$
  - eff: $R = y$
  - cost: 1
- $movePadlock(x, y)$ – move from x to y, in case of unlocked padlock; $(x, y) \in \{(3, 4), (4, 3), (4, 5), (5, 4)\}$
  - pre: $R = x, P = 0$
  - eff: $R = y$
  - cost: 1
- $pickUpKeyA$
  - pre: $R = 2, K_A = 0$
  - eff: $K_A = 1$
  - cost: 1
- $pickUpKeyB$
  - pre: $R = 7, K_B = 0$
  - eff: $K_B = 1$
  - cost: 1
- $unlockPadlock(x)$ – $x \in \{3, 4\}$
  - pre: $R = x, K_A = 1$
  - eff: $P = 0$
  - cost: 1

1.2. Write down **FDR encoding** of the **Meeting Point** problem from Exercise Sheet 1, Exercise 1.3:

Friends Alice, Emil, and Frank want to meet somewhere in the city. Find the meeting point and route for each friend so that the overall travelled distance is minimal. Figure 2 shows the map of the city with distances between locations, Alice live at location A, Emil at location E, and Frank at location F.

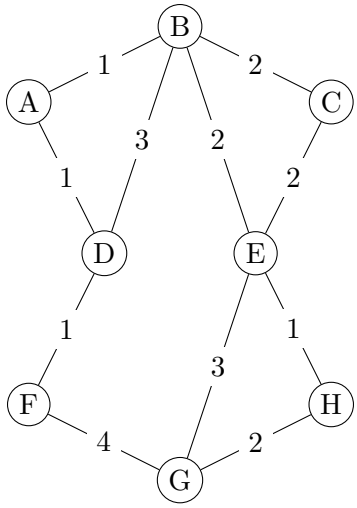

Figure 2: City map with distances between locations.

(a) Start with variables and their domains.

(b) Then write action encodings. You do not need to enumerate all of them one by one, instead write them in some schematic way and do not forget to clearly state what are preconditions and effects in terms of variable assignements and what are costs of the actions.

(c) Then specify the intial state.

(d) Lastly, and this is the hardest part, think about how to specify the goal states. Remember that in FDR we are allowed to specify only a single partial state, but the friends can meet at any location.

Solution:

Variables:

- a, e, f, $D_a = D_e = D_f = \{A, ..., H\}$ – positions of friends
- g, $D_g = \{0, 1\}$ – goal achieved

Initial state: $\{a{=}A, e{=}E, f{=}F, g{=}0\}$

Goal: $\{g{=}1\}$

Actions:

- $move(x, y, z)$ – move friend x from y to z; $x \in \{a, e, f\}$, (y, z) according to the connections in the map
  - pre: $x = y$
  - eff: $x = z$
  - cost: according to the map
- $achieveGoal(x)$ – model disjunctive goal; $x \in \{A, ..., H\}$
  - pre: $a = x, e = x, f = x$
  - eff: $g = 1$
  - cost: 0 – we do not want this to count to the cost of the solution
- To further model that no other state is reachable after reaching the goal, we can add $g = 0$ as a precondition to the $move()$ actions.

1.3. Consider the following **Distaster Relief** problem. A natural disaster (earthquake, flood, hurricane) has struck a region. Civilians in several locations require evacuation to safe shelters, and some are injured and must be transported to a hospital. A team of rescue vehicles must coordinate to transport people under limited time and resource constraints.

We have locations: depot (D), two villages (V1 and V2), hospital (H), and shelter (S). We have one truck (T) that can carry healthy people, and one ambulance (A) that can carry only injured people. The truck T has a capacity for carrying 5 people. The ambulance can carry at most 2 injured people.

Initially, V1 needs to evacuate 6 people to the shelter S, and transport 3 people to the hospital H. V2 needs to evacuate 8 people to S, and transport 2 people to H. The truck and ambulance are initially empty and parked in the depot D. The travel times between locations are shown in Table 1 (it is symmetric).

The goal is evacuate to the shelter and transport to the hospital all people that need it, and minimize overall time of achieving it, assuming the time for pick-up and drop-out of people is negligible in comparison to travel times.

	D	V1	V2	H	S
D	–	2	4	3	5
V1	2	–	3	2	4
V2	4	3	–	3	2
H	3	2	3	–	3
S	5	4	2	3	–

Table 1: Road travel times between locations (in hours).

Model this problem in FDR.

Solution:

Variables:

- T, $D_T = \{D, V1, V2, H, S\}$ – location of the truck
- A, $D_A = \{D, V1, V2, H, S\}$ – location of the ambulance
- P_T, $D_{P_T} = \{0, 1, \ldots, 5\}$ – number of people carried by the truck
- P_A, $D_{P_A} = \{0, 1, 2\}$ – number of people carried by the ambulance
- E_{V1}, E_{V2}, $D_{E_{V1}} = D_{E_{V2}} = \{0, 1, \ldots, 8\}$ – number of people still waiting for the evacuation in V1 and V2, respectively

- I_{V1}, I_{V2}, $D_{I_{V1}} = D_{I_{V2}} = \{0, 1, \ldots, 3\}$ – number of injured people still waiting for the transport in V1 and V2, respectively

Initial state: $\{T{=}D, A{=}D, S_T{=}0, P_T{=}0, P_A{=}0, E_{V1}{=}6, E_{V2}{=}8, I_{V1}{=}3, I_{V2}{=}2\}$

Goal: $\{E_{V1}{=}0, E_{V2}{=}0, I_{V1}{=}0, I_{V2}{=}0, P_T = 0, P_A = 0\}$

Actions:

- $move(x, y, z)$ – move $x \in \{T, A\}$ from $y \in \{D, V1, V2, H, S\}$ to $z \in \{D, V1, V2, H, S\}$
  - pre: $x = y$
  - eff: $x = z$
  - cost: according to the table above
- $pickUpForEvacuation(l, b, a, b', a')$ – pick up a person to T at location l; b, a and b', a' keep track of capacities of truck/locations before and after application of this action; $l \in \{V1, V2\}$, $(b, a) \in \{(0, 1), (1, 2), \ldots, (4, 5)\}$, $(b', a') \in \{(8, 7), (7, 6), \ldots, (1, 0)\}$
  - pre: $T = l, P_T = b, E_l = b'$
  - eff: $P_T = a, E_l = a'$
  - cost: 0
- $dropAtShelter(b, a)$ – drop a person from T at the shelter S; $(b, a) \in \{(5, 4), (4, 3), \ldots, (1, 0)\}$
  - pre: $T = S, P_T = b$
  - eff: $P_T = a$
  - cost: 0
- $pickUpForHospital(l, b, a, b', a')$ – as above but for injured people; $l \in \{V1, V2\}$, $(b, a) \in \{(0, 1), (1, 2)\}$, $(b', a') \in \{(3, 2), (2, 1), (1, 0)\}$
  - pre: $A = l, P_A = b, I_l = b'$
  - eff: $P_A = a, I_l = a'$
  - cost: 0
- $dropAtHospital(b, a)$ – as above but for injured people; $(b, a) \in \{(2, 1), (1, 0)\}$
  - pre: $A = H, P_A = b$
  - eff: $P_A = a$
  - cost: 0

2 Delete Relaxation

2.1. Go back to your solution to Exercise 1.1 and list all delete-relaxed states *reachable* from the initial state. Further, list 2 states that are not reachable under delete relaxation.

Solution:

Not delete-relax reachable is, for example, any superset of $\{R{=}5, K_A{=}0\}$ because we cannot step to $4, ..., 7$ without holding the key A. Similarly any superset of $\{R{=}5, P{=}1\}$ because the padlock needs to be unlocked for us to move to 4 and further.

The states are

$\{R{=}1, K_A{=}0, K_B{=}0, P{=}1\}$,

$\{R{=}1, R{=}2, K_A{=}0, K_B{=}0, P{=}1\}$,

$\{R{=}1, R{=}2, R{=}3, K_A{=}0, K_B{=}0, P{=}1\}$,

$\{R{=}1, R{=}2, K_A{=}0, K_A{=}1, K_B{=}0, P{=}1\}$,

$\{R{=}1, R{=}2, R{=}3, K_A{=}0, K_A{=}1, K_B{=}0, P{=}1\}$,

$\{R{=}1, R{=}2, R{=}3, K_A{=}0, K_A{=}1, K_B{=}0, P{=}1, P{=}0\}$,

$\{R{=}1, R{=}2, R{=}3, R{=}4, K_A{=}0, K_A{=}1, K_B{=}0, P{=}1, P{=}0\}$,

$\{R{=}1, R{=}2, R{=}3, R{=}4, R{=}5, K_A{=}0, K_A{=}1, K_B{=}0, P{=}1, P{=}0\}$,

$\{R{=}1, R{=}2, R{=}3, R{=}4, R{=}5, R{=}6, K_A{=}0, K_A{=}1, K_B{=}0, P{=}1, P{=}0\}$,

$\{R{=}1, R{=}2, R{=}3, R{=}4, R{=}5, R{=}6, R{=}7, K_A{=}0, K_A{=}1, K_B{=}0, P{=}1, P{=}0\}$,

$\{R{=}1, R{=}2, R{=}3, R{=}4, R{=}5, R{=}6, R{=}7, K_A{=}0, K_A{=}1, K_B{=}0, K_B{=}1, P{=}1, P{=}0\}$,

2.2. Consider the following planning task where you are at home and you want to eat an ice-cream. Your home is in the woods and the only way to get to the city and back is by taking a bus that cost one euro per ride. The ice-cream costs also 1 euro and you only have 2 euros left (it's the end of the month).

- $\mathcal{V} = \{\textit{have-icecream}, \textit{at}, \textit{money}\}$, with
  - $D_{\textit{have-icecream}} = \{N, Y\}$,
  - $D_{\textit{at}} = \{H, C\}$.
  - $D_{\textit{money}} = \{0, 1, 2\}$.
- $\mathcal{A} = \{\textit{go-by-bus}(x, y, m), \textit{buy-icecream}(m)\}$, with
  - $\textit{go-by-bus}(x, y, m)$ for $(x, y) \in \{(H, C), (C, H)\}$ and $m \in \{1, 2\}$
    pre:$\{\textit{at} = x, \textit{money} = m\}$,
    eff:$\{\textit{at} = y, \textit{money} = m - 1\}$
    cost: 1,
  - $\textit{buy-icecream}(m)$ for $m \in \{1, 2\}$
    pre:$\{\textit{at} = C, \textit{money} = m\}$,
    eff:$\{\textit{have-icecream} = Y, \textit{money} = m - 1\}$
    cost: 1.
- $I = \{\textit{money} = 2, \textit{at} = H, \textit{have-icecream} = N\}$.
- $G = \{\textit{at} = H, \textit{have-icecream} = Y\}$.

(a) Compute the value of $h^*(I)$

Solution: ∞

(b) Compute the value of $h^+(I)$

Solution: 2

(c) Compute heuristic value using `SimpleDeleteRelaxationHeur`.

Solution: 2

2.3. Consider the following **Elevator** FDR planning task.

- $\mathcal{V} = \{e, l, p_1, p_2, p_3\}$, $D_e = \{1, 2, 3\}$, $D_l = \{0, 1, 2\}$, $D_{p_i} = \{1, 2, 3, e\}$ for $i \in \{1, 2, 3\}$
  - e: current floor of the elevator
  - l: number of people in the elevator
  - p_i: location of person i (on the floor or inside the elevator)
- $I = \{e{=}1, l{=}0, p_1{=}1, p_2{=}2, p_3{=}3\}$
- $G = \{p_1{=}3, p_2{=}1, p_3{=}2\}$
- $\mathcal{A} = \{moveUp(x), moveDown(x), getIn(x, y, z), getOut(x, y, z)\}$ where
  - $moveUp(x)$ for $x \in \{1, 2\}$
    - pre: $e{=}x$
    - eff: $e{=}x + 1$
    - cost: 10
  - $moveDown(x)$ for $x \in \{2, 3\}$
    - pre: $e{=}x$
    - eff: $e{=}x - 1$
    - cost: 10
  - $getIn(x, y, z)$ for $x \in \{1, 2, 3\}$, $y \in \{1, 2, 3\}$, $z \in \{0, 1\}$
    - pre: $e{=}x, p_y{=}x, l{=}z$
    - eff: $p_y{=}e, l{=}z + 1$
    - cost: 1
  - $getOut(x, y, z)$ for $x \in \{1, 2, 3\}$, $y \in \{1, 2, 3\}$, $z \in \{1, 2\}$
    - pre: $e{=}x, p_y{=}e, l{=}z$
    - eff: $p_y{=}x, l{=}z - 1$
    - cost: 1

(a) Write down an optimal plan and $h^*(I)$.

Solution:
Optimal plan (there are others):
$(getIn(1, 1, 0), moveUp(1), getIn(2, 2, 1), moveUp(2), getOut(3, 1, 2),$
$getIn(3, 3, 1), moveDown(3), getOut(2, 3, 2), moveDown(2), getOut(1, 2, 1))$
$h^*(I) = 46$

(b) Compute $h^*(s)$ for the state $s = \{e{=}2, l{=}0, p_1{=}3, p_2{=}2, p_3{=}2\}$.

Solution:
Optimal s-plan: $(getIn(2, 2, 0), moveDown(2), getOut(1, 2, 1))$
$h^*(s) = 12$

(c) Write down optimal delete-relaxed plan and $h^+(I)$.

Solution:
Optimal delete-relaxed plan (there are others):
$(moveUp(1), moveUp(2), getIn(1,1,0), getIn(2,2,0), getIn(3,3,0),$
$getOut(3,1,1), getOut(1,2,1), getOut(2,3,1))$
$h^+(I) = 26$

(d) Write down optimal delete-relaxed plan for the state s defined above.

Solution:
Optimal s-plan: $(getIn(2,2,0), moveDown(2), getOut(1,2,1))$
$h^*(s) = 12$

(e) Run `SimpleDeleteRelaxationHeur` on the initial state. Write down content of A', F and C at the end of each cycle. What is the resulting heuristic estimate?

Solution:
For column F, only added facts are shown.

Step	F	C	A'
0.	$e{=}1, l{=}0, p_1{=}1, p_2{=}2, p_3{=}3$	0	
1.	$e{=}2, l{=}1, p_1{=}e$	1	$getIn(1,1,0), moveUp(1)$
2.	$e{=}3, l{=}2, p_2{=}e, p_1{=}2$	1	$getIn(2,2,0), getIn(2,2,1), moveUp(2)$ $moveDown(2), getIn(1,1,1), getOut(2,1,1)$ $getOut(1,1,1)$
3.	$p_3{=}e, p_1{=}3, p_2{=}1, p_2{=}3$	1	$getIn(3,3,0), getIn(3,3,1), moveDown(3)$ $getIn(2,1,0), getIn(2,1,1), getOut(1,1,2)$ $getOut(2,1,2), getOut(3,1,1), getOut(3,1,2)$ $getOut(1,2,1), getOut(1,2,2), getOut(2,2,1)$ $getOut(2,2,2), getOut(3,2,1), getOut(3,2,2)$
4.	$p_3{=}1, p_3{=}2$	1	$getOut(1,3,1), getOut(1,3,2), getOut(2,3,1)$ $getOut(2,3,2), getOut(3,3,1), getOut(3,3,2)$ $getIn(1,2,0), getIn(1,2,1)$

Heuristic estimate: 4

3 FF Heuristic

3.1. Compute FF heuristic for the initial state of the problem from 2.2.

Write down all generated fact and action layers (F_i and A_i) and mark selected actions (you can as well mark all subgoals if you want to).

Solution:

$F_0 = \{money{=}2, at{=}H, have\text{-}icecream{=}N\}$

$A_0 = \{go\text{-}by\text{-}bus(H, C, 2)\}$

$F_1 = F_0 \cup \{money{=}1, at{=}C\}$

$A_1 = A_0 \cup \{go\text{-}by\text{-}bus(C, H, 2), go\text{-}by\text{-}bus(H, C, 1), go\text{-}by\text{-}bus(C, H, 1), buy\text{-}icecream(2), buy\text{-}icecream(1)$

$F_2 = F_0 \cup \{money{=}0, have\text{-}icecream{=}Y\}$

Heuristic value: 2

3.2. Go back to your solution to Exercise 1.1 and compute FF heuristic for (a) the initial state, and (b) the state where robot is at cell 3 and is holding the key A.

For each (a) and (b), write down all generated fact and action layers (F_i and A_i) and mark selected actions (you can as well mark all subgoals if you want to).

Solution:

(a)

$F_0 = \{R{=}1, K_A{=}0, K_B{=}0, P{=}1\}$

$A_0 = \{move(1,2)\}$

$F_1 = F_0 \cup \{R{=}2\}$

$A_1 = A_0 \cup \{move(2,3), pickUpKeyA, move(2,1)\}$

$F_2 = F_1 \cup \{R{=}3, K_A{=}1\}$

$A_2 = A_1 \cup \{unlockPadlock, move(3,2)\}$

$F_3 = F_2 \cup \{P{=}0\}$

$A_3 = A_2 \cup \{movePadlock(3,4)\}$

$F_4 = F_3 \cup \{R{=}4\}$

$A_4 = A_3 \cup \{movePadlock(4,5), movePadlock(4,3)\}$

$F_5 = F_4 \cup \{R{=}5\}$

$A_5 = A_4 \cup \{move(5,6), movePadlock(5,4)\}$

$F_6 = F_5 \cup \{R{=}6\}$

$A_6 = A_5 \cup \{move(6,7), move(6,5)\}$

$F_7 = F_6 \cup \{R{=}7\}$

$A_7 = A_6 \cup \{pickUpKeyB, move(7,6)\}$

$F_8 = F_7 \cup \{K_B{=}1\}$

Heuristic value: 9

(b)

$F_0 = \{R{=}3, K_A{=}1, K_B{=}0, P{=}1\}$

$A_0 = \{unlockPadlock, move(3,2)\}$

$F_1 = F_0 \cup \{P{=}0, R{=}2\}$

$A_1 = A_0 \cup \{movePadlock(3,4), move(2,3), move(2,1)\}$

$F_2 = F_1 \cup \{R{=}4, R{=}1\}$

$A_2 = A_1 \cup \{movePadlock(4,5), movePadlock(4,3)\}$

$F_3 = F_2 \cup \{R{=}5\}$

$A_3 = A_2 \cup \{move(5,6), movePadlock(5,4)\}$

$F_4 = F_3 \cup \{R{=}6\}$

$A_4 = A_3 \cup \{move(6,7), move(6,5)\}$

$F_5 = F_4 \cup \{R{=}7\}$

$A_5 = A_4 \cup \{pickUpKeyB, move(7,6)\}$

$F_6 = F_5 \cup \{K_B{=}1\}$

Heuristic value: 6

3.3. Compute FF heuristic for the initial state of the problem from 2.3.

Write down all generated fact and action layers (F_i and A_i) and mark selected actions (you can as well mark all subgoals if you want to).

Solution:

There are different solutions (with different h-values)!

$F_0 = \{e{=}1, l{=}0, p_1{=}1, p_2{=}2, p_3{=}3\}$

$A_0 = \{getIn(1,1,0), moveUp(1)\}$

$F_1 = F_0 \cup \{e{=}2, l{=}1, p_1{=}e\}$

$A_1 = A_0 \cup \{getIn(2,2,0), getIn(2,2,1), moveUp(2),$
$moveDown(2), getIn(1,1,1), getOut(2,1,1), getOut(1,1,1)\}$

$F2 = F_1 \cup \{e{=}3, l{=}2, p_2{=}e, p_1{=}2\}$

$A_2 = A_1 \cup \{getIn(3,3,0), getIn(3,3,1), moveDown(3),$
$getIn(2,1,0), getIn(2,1,1), getOut(1,1,2),$
$getOut(2,1,2), getOut(3,1,1), getOut(3,1,2),$
$getOut(1,2,1), getOut(1,2,2), getOut(2,2,1),$
$getOut(2,2,2), getOut(3,2,1), getOut(3,2,2)\}$

$F_3 = F_2 \cup \{p_3{=}e, p_1{=}3, p_2{=}1, p_2{=}3\}$

$A_3 = A_2 \cup \{getOut(1,3,1), getOut(1,3,2), getOut(2,3,1),$
$getOut(2,3,2), getOut(3,3,1), getOut(3,3,2),$
$getIn(1,2,0), getIn(1,2,1)\}$

$F_4 = F_3 \cup \{p_3{=}1, p_3{=}2\}$

Heuristic estimate: 27 – compare to h^+...